Forensic Analysis and Recovery of Corrupted GPT Disk Structures

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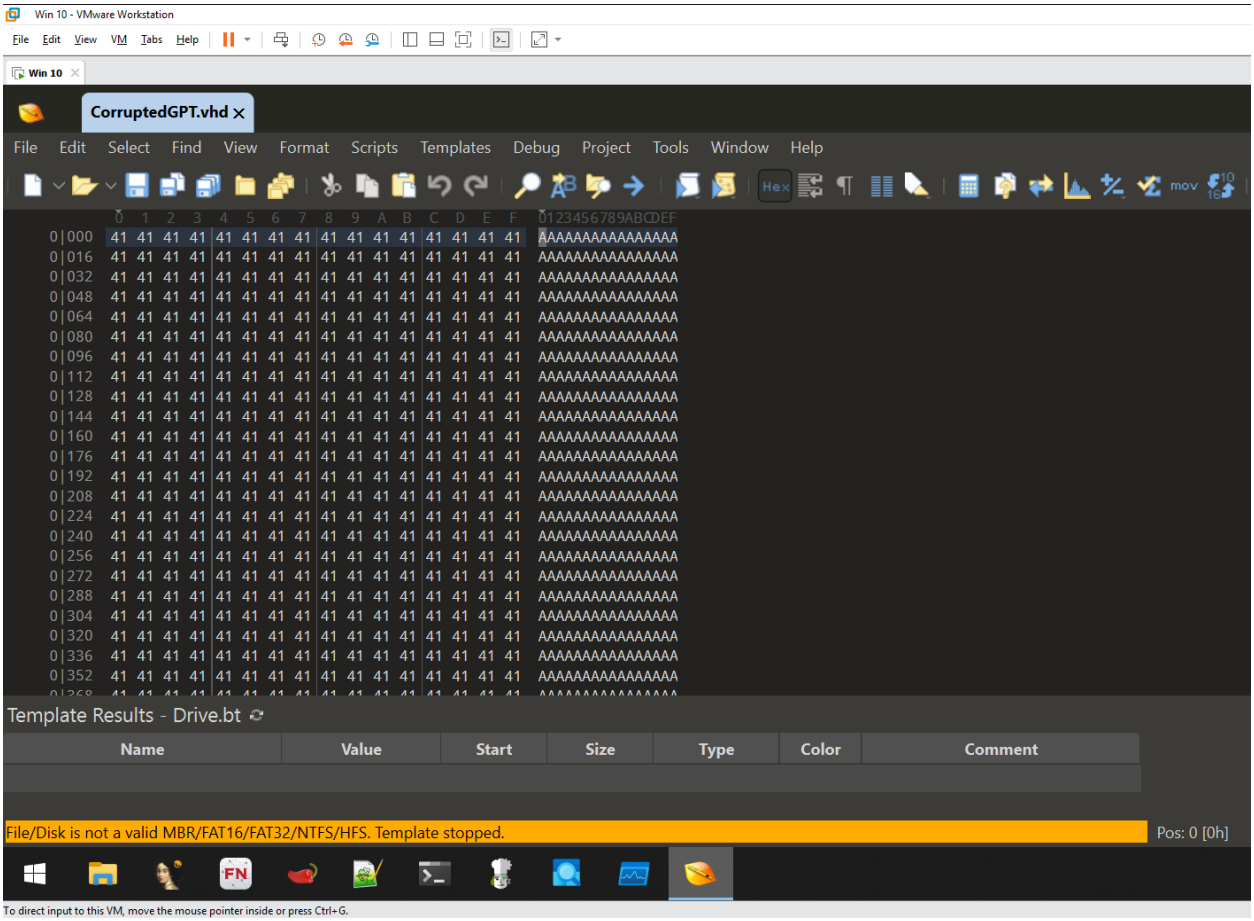
Tools Used: 010 Editor, FTK Imager

1. Incident Description

In this scenario, a GPT-formatted virtual disk was intentionally damaged by overwriting both:

- the **Protective Master Boot Record (Protective MBR)** located in sector 0
- the **Primary GPT Header** located in sector 1



2. GPT Disk Layout

The primary GPT structure at the beginning of the disk contains:

Sector	Structure	Purpose
Sector 0	Protective MBR	Protects GPT disks from legacy MBR tools
Sector 1	Primary GPT Header	Stores GPT metadata
Sector 2–33	Primary Partition Entry Array	Stores partition definitions

The backup GPT structure is stored at the end of the disk and contains:

Location	Structure	Purpose
Last 33 sectors	Backup Partition Entry Array + Backup GPT Header	Provides redundancy for recovery

This dual header design allows recovery even when the primary GPT metadata is damaged.

In this case, both the Protective MBR and Primary GPT Header were corrupted, making the backup GPT structures essential for reconstruction.

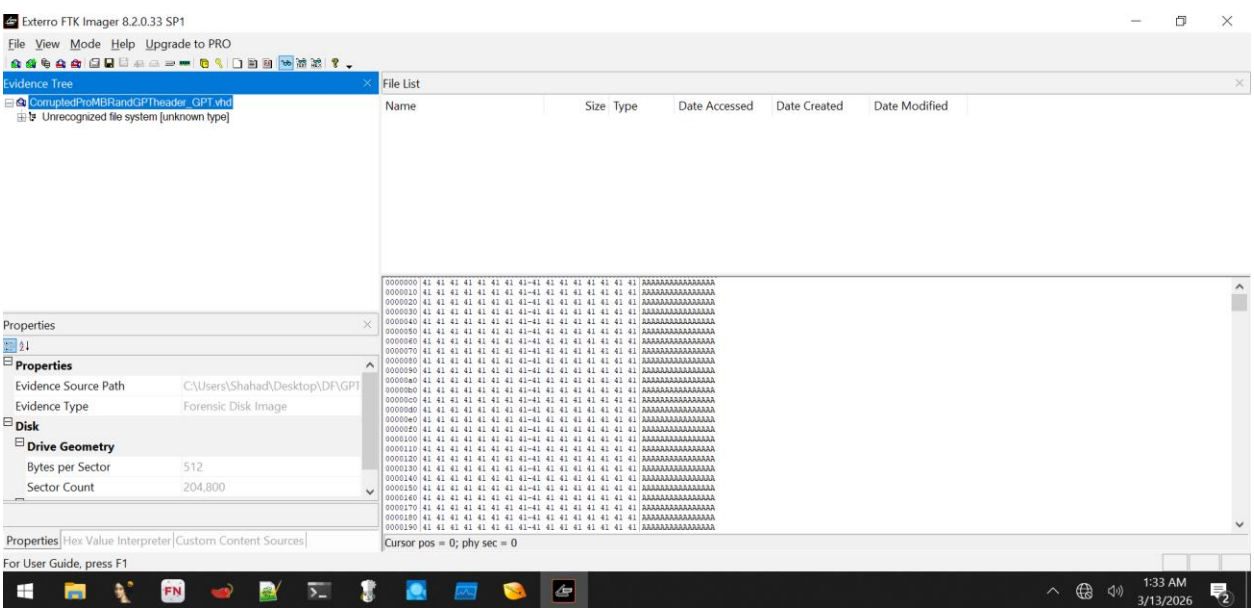
3. Initial Examination

The damaged disk image was opened using FTK Imager.

Initial result showed:

- disk visible
- no valid partitions recognized
- file system reported as unknown

This confirmed that the primary GPT structures were no longer readable.



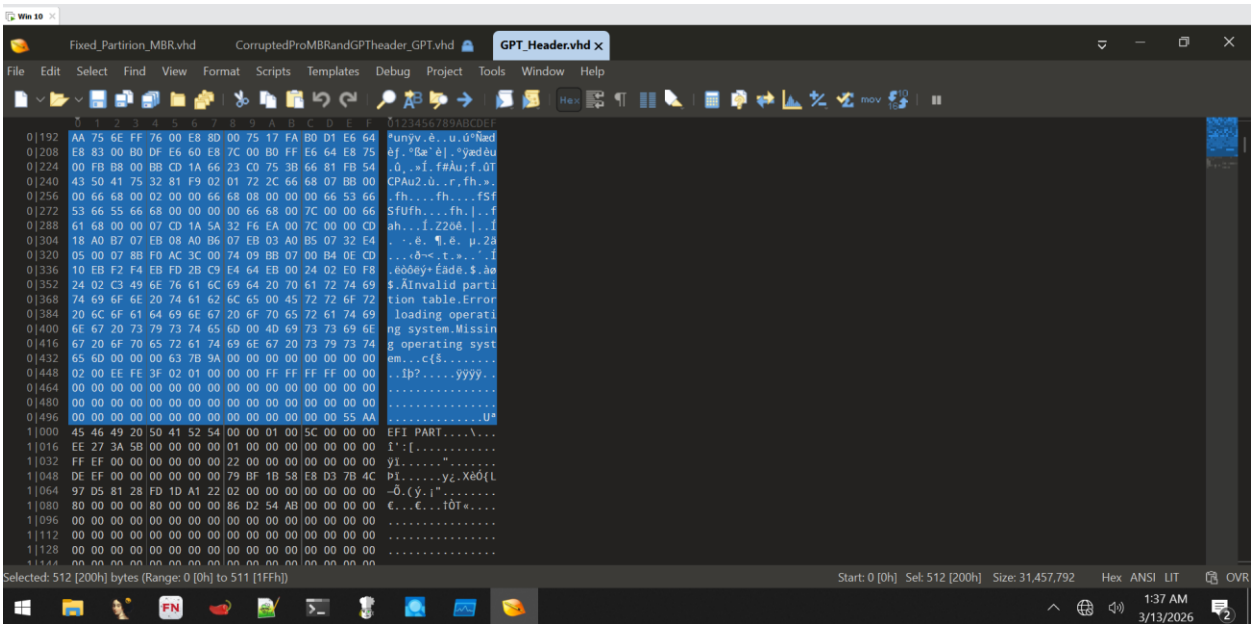
4. Recovery of the Protective MBR

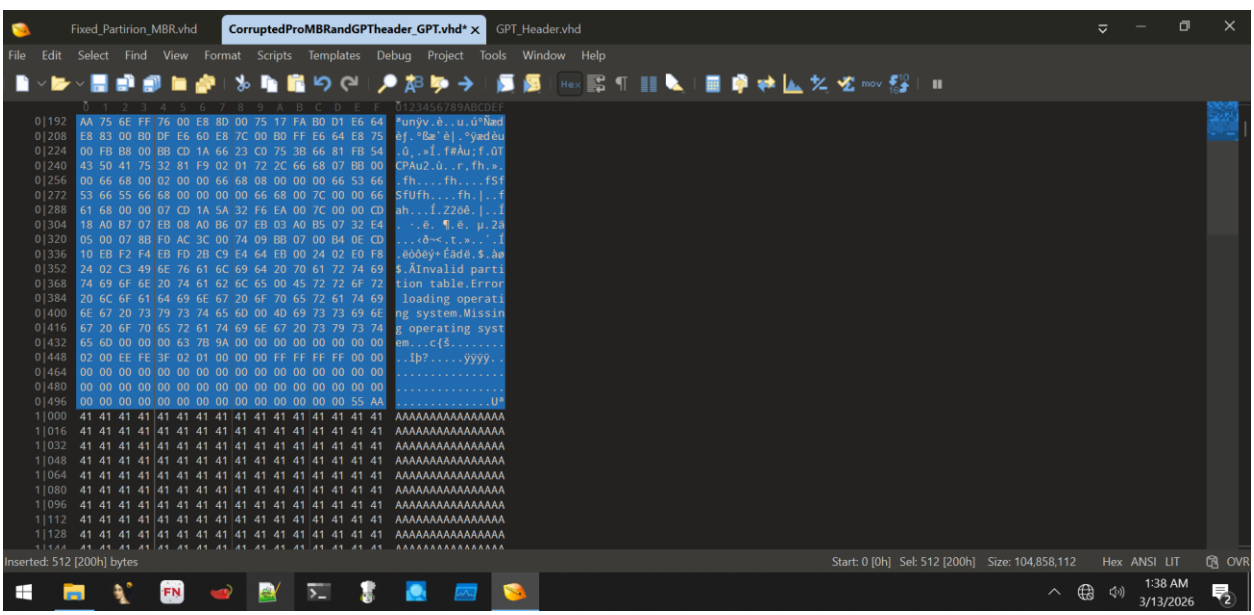
Because the original Protective MBR located in sector 0 was corrupted a new GPT disk with the same basic layout was created and opened in 010 Editor.

A full **512-byte sector** was copied from the clean GPT disk and used to replace the damaged first sector of the corrupted disk.

The copied sector included:

- boot code area
- protective partition entry
- boot signature 55 AA





5. Recovery of the Primary GPT Header

After restoring the Protective MBR, the damaged primary GPT header remained corrupted.

GPT provides built-in redundancy by storing a backup copy of the GPT header at the end of the disk.

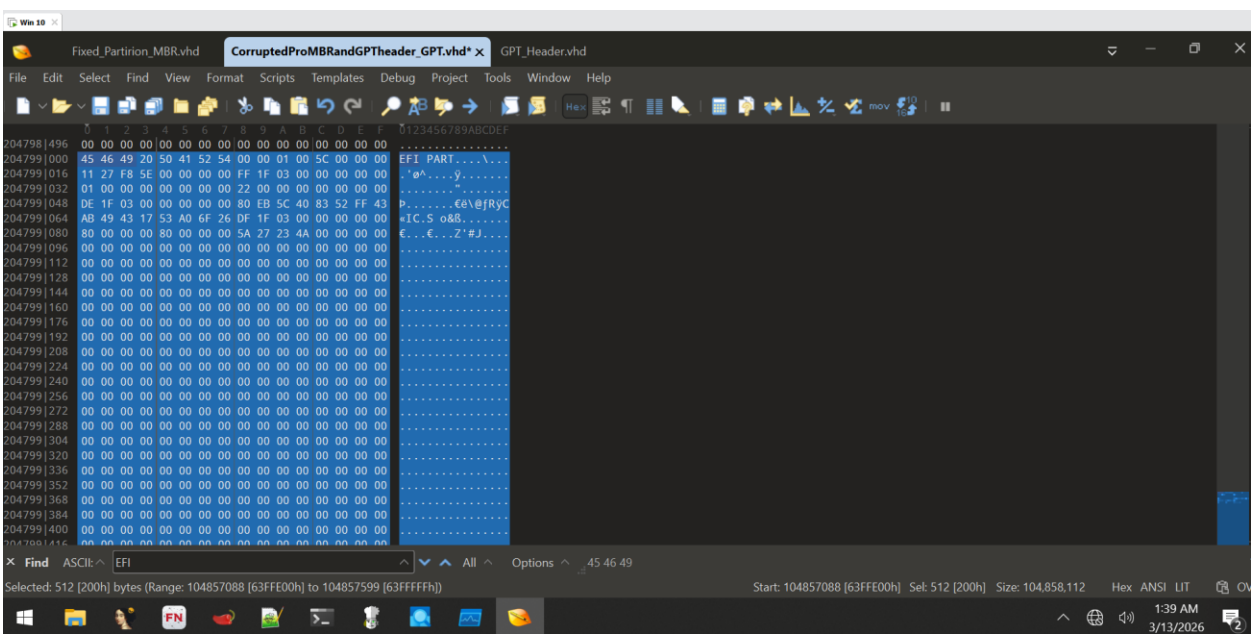
To locate this backup header, the disk was searched for the GPT signature: **EFI PART**

A full **512-byte backup header** was copied from the last sector and used to replace the damaged primary GPT header located at sector 1.

This restored the GPT header structure, including:

- GPT signature
- header size
- CRC information
- LBA references
- partition entry location

Because the backup header contains values relative to the end of the disk, several fields required manual correction before final validation.



6. Locating the Backup Partition Entry Array

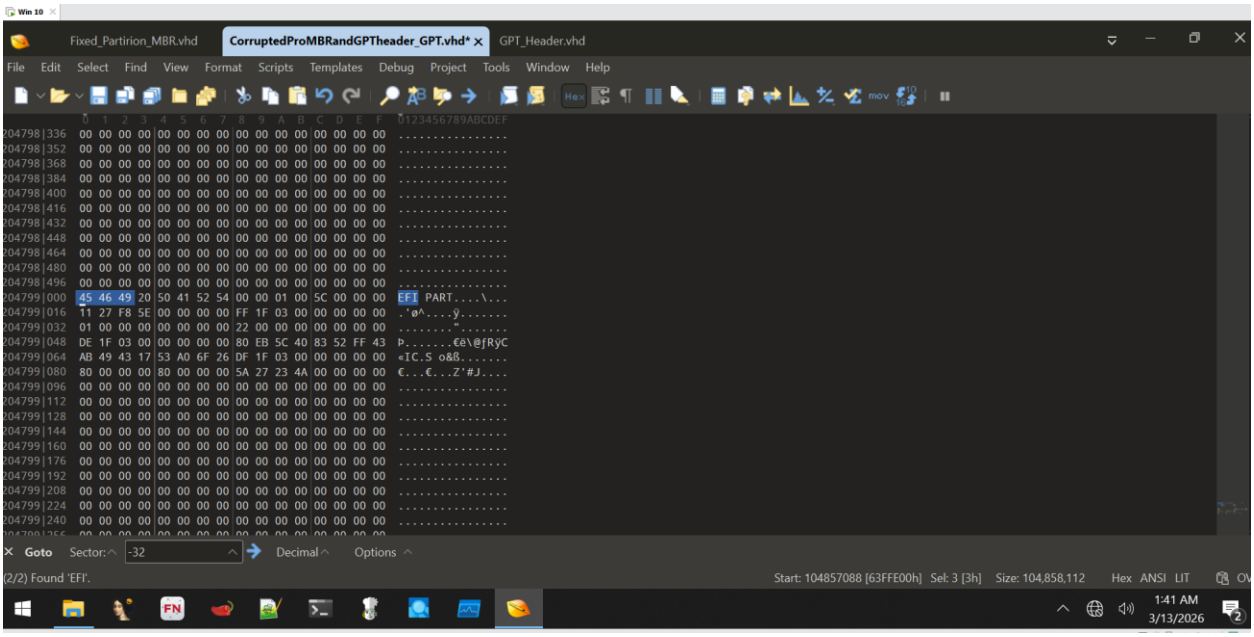
After identifying the backup GPT header at the end of the disk, the next step was to locate the backup partition entry information.

In GPT structure, the backup partition entry array is stored immediately before the backup GPT header.

Because the partition entry array occupies **32 sectors**, the location was determined by moving backward from the backup GPT header by: **32 sectors**

This was performed in 010 Editor using sector navigation.

Thus: **Backup GPT Header sector − 32 sectors = Backup Partition Entry Array**



After locating the backup partition entry array, the required partition descriptors were recovered from the backup metadata.

A total of **256 bytes** were copied from the backup partition entry area and written into the damaged primary partition entry location.

This size corresponds to:

- 2 GPT partition entries
- each GPT partition entry = 128 bytes

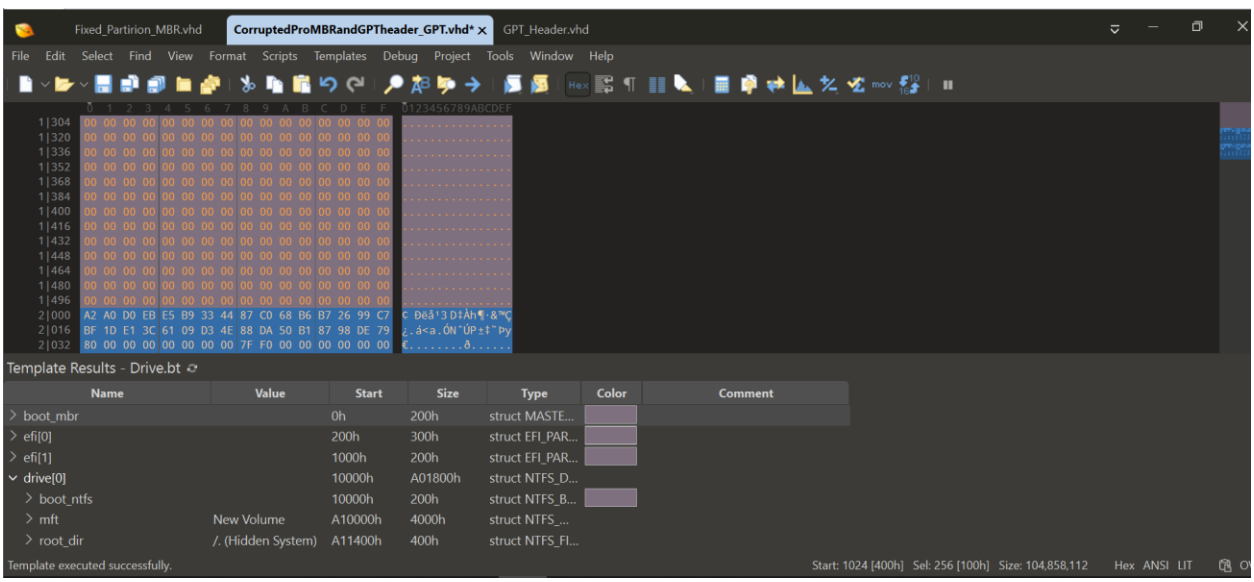
Therefore: 2 times128=256

```
204766|464  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
204766|480  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
204766|496  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
204767|000  A2 A0 D0 EB E5 B9 33 44 87 C0 68 B6 B7 26 99 C7  ¢ Ðëå¹3D‡Àh¶·&™Ç
204767|016  BF 1D E1 3C 61 09 D3 4E 88 DA 50 B1 87 98 DE 79  ¿.á<a.ÓN^ÚP±‡˜Þy
204767|032  80 00 00 00 00 00 00 00 7F F0 00 00 00 00 00 00  €........ð......
204767|048  00 00 00 00 00 00 00 00 42 00 61 00 73 00 69 00  ........B.a.s.i.
204767|064  63 00 20 00 64 00 61 00 74 00 61 00 20 00 70 00  c. .d.a.t.a. .p.
204767|080  61 00 72 00 74 00 69 00 74 00 69 00 6F 00 6E 00  a.r.t.i.t.i.o.n.
204767|096  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
204767|112  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
204767|128  A2 A0 D0 EB E5 B9 33 44 87 C0 68 B6 B7 26 99 C7  ¢ Ðëå¹3D‡Àh¶·&™Ç
204767|144  A5 39 B6 08 C6 4C 7B 49 A5 02 AD F1 15 A1 B4 9B  ¥9¶.ÆL{I¥.-ñ.¡´›
204767|160  80 F0 00 00 00 00 00 00 7F E0 01 00 00 00 00 00  €ð.......à......
204767|176  00 00 00 00 00 00 00 00 42 00 61 00 73 00 69 00  ........B.a.s.i.
204767|192  63 00 20 00 64 00 61 00 74 00 61 00 20 00 70 00  c. .d.a.t.a. .p.
204767|208  61 00 72 00 74 00 69 00 74 00 69 00 6F 00 6E 00  a.r.t.i.t.i.o.n.
204767|224  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
204767|240  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
```

7. Manual Correction of GPT Header Fields

After copying the backup GPT header to sector 1, several header fields still reflected backup-header values and required manual correction before the disk could function correctly.

Because the copied header originated from the final sector of the disk, the logical addresses inside the header were reversed relative to the primary GPT structure.



7.1 Correcting CurrentLBA

The backup header originally showed:

CurrentLBA = 31FFFh

This value corresponds to the last sector of the disk because the backup GPT header is stored at the end of the disk.

For a valid primary GPT header, the field was corrected to: **CurrentLBA = 1h**

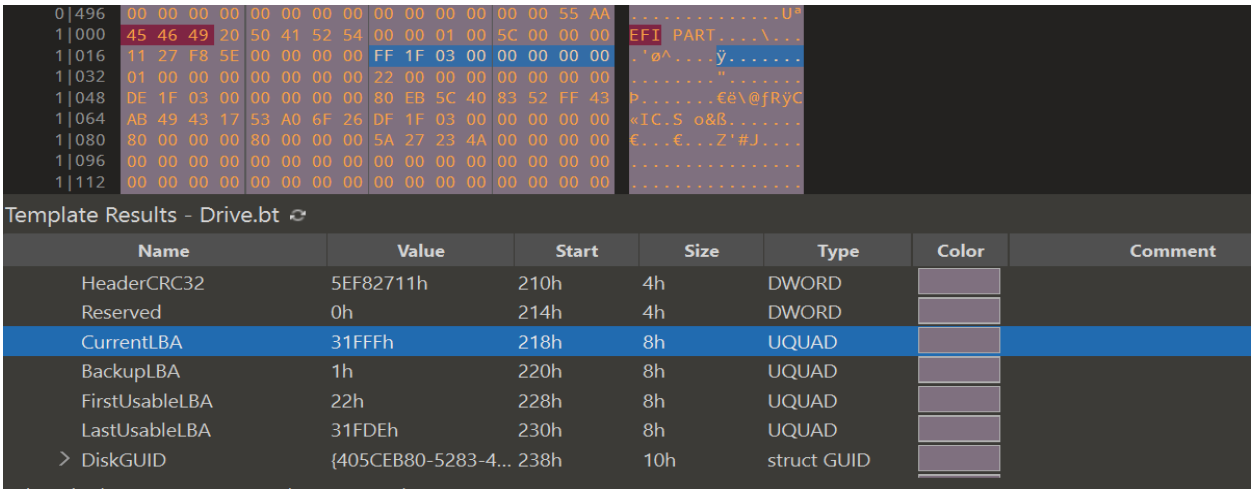
7.2 Correcting BackupLBA

The backup header originally contained: **BackupLBA = 1h**

This value represents the location of the primary GPT header.

For the restored primary header, BackupLBA must reference the backup header stored at the last sector: **BackupLBA = 31FFFh**

Thus, the two values were swapped.



7.3 Correcting Partition Entry Starting Location

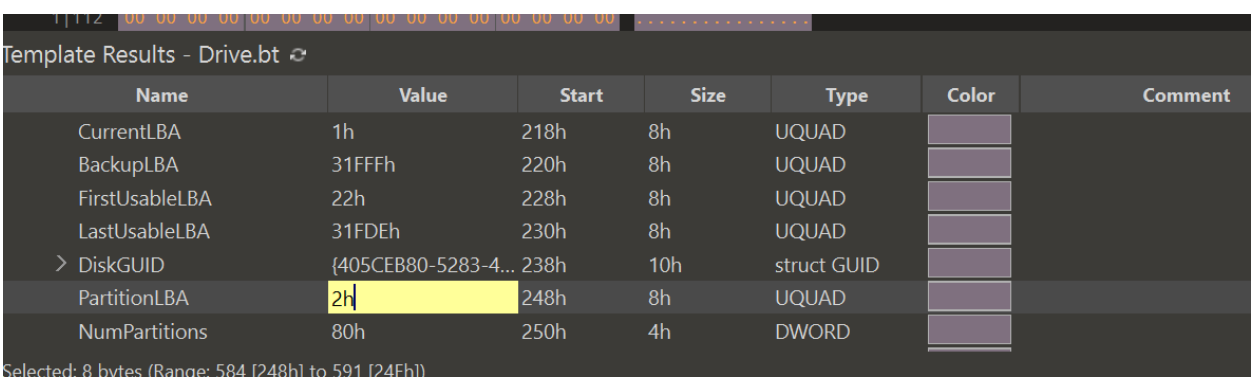
The field "**PartitionLBA**" defines where GPT partition entries begin.

The corrected value was: **PartitionLBA = 2h**

This is correct because GPT structure reserves:

- sector 0: Protective MBR
- sector 1: Primary GPT Header
- sector 2: Partition Entry Array

Thus: **Partition entries begin immediately at sector 2**



8. Verification of GPT Repair and Evidence Recovery

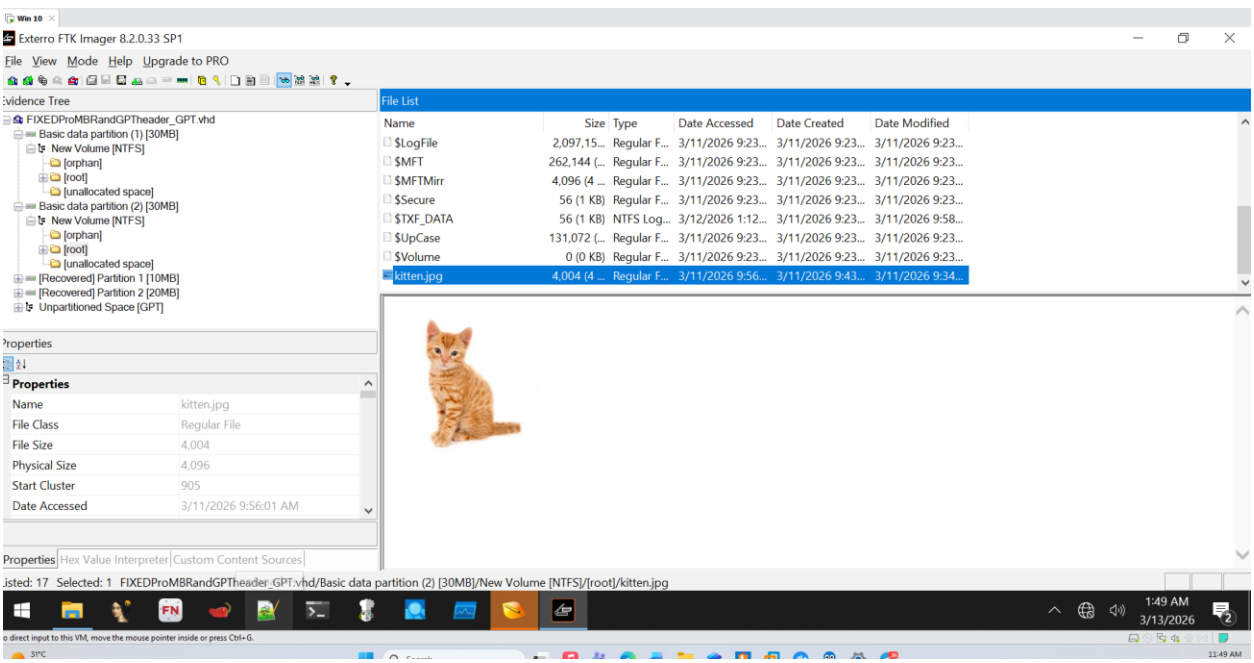
After completing the manual correction of GPT header fields and restoring the required partition entries, the repaired disk image was reopened using FTK Imager to verify whether the disk structure could be interpreted correctly.

FTK Imager successfully recognized the restored GPT layout and displayed both NTFS partitions as valid accessible volumes.

This confirmed that:

- the Protective MBR was valid
- the primary GPT header was correctly restored
- partition entries were readable
- NTFS file systems remained intact

Within the recovered NTFS partition, previously inaccessible evidence became available again.



Recovered files included: **kitten.jpg**

The image opened successfully inside FTK Imager, confirming full file-level recovery.